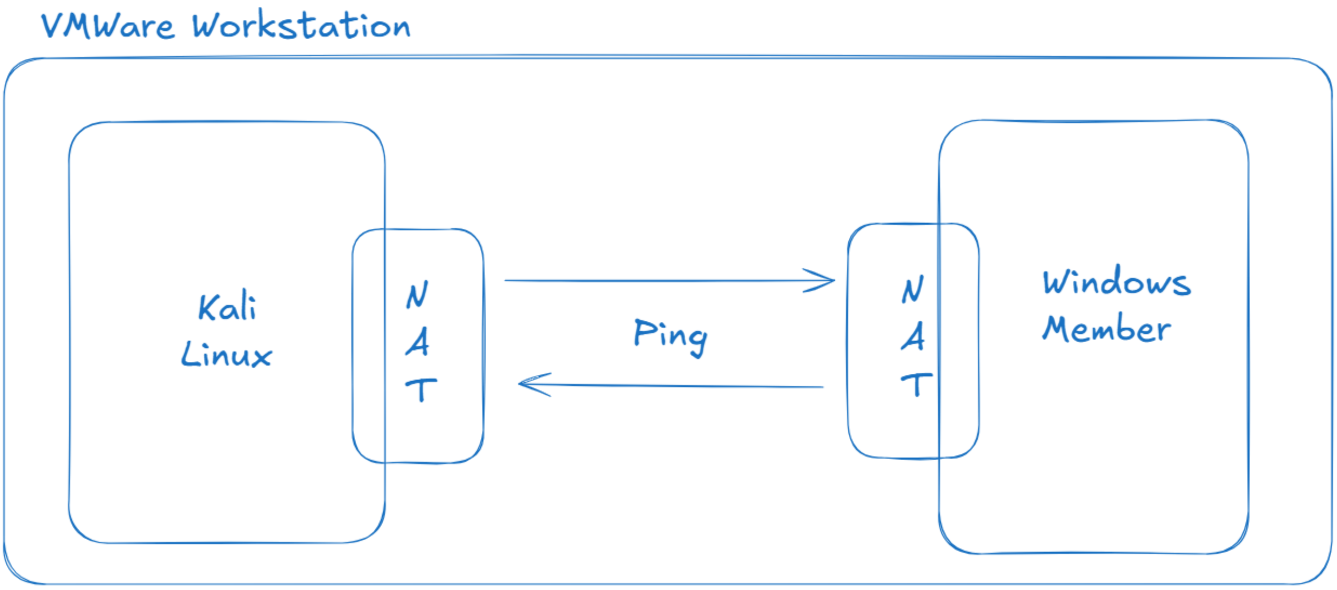
- Students to analyse the given malware analysis link and answer the questions
    - Lab setup:
        - 1st VM: Kali linux
            - Tools needed:
                - strings
                - sha256sum
                - tcpdump
                - python3 -m http.server
        - 2nd VM: Windows Server 2022 (member)
            - Tools needed:
                - Windows Defender
                - PowerShell
                - Event Viewer

Labsetup:



On kali linux:

Create a Test "Malware" File (EICAR - European Institute for Computer Anti-virus Research (EICAR))

- mkdir ~/malware-demo
- cd ~/malware-demo
- echo *'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*'* > eicar.txt
    - This is 68-bytes string of ASCII text.
    - This is a fixed string.
    - EICAR is used for testing anti-virus software functionality without using real Malware.

Verify hash:

- sha256sum eicar.txt

Start a Local Web Server (Delivery Simulation)

- python3 -m http.server 8080

```
┌──(root㉿kali)-[~]
└─# mkdir ~/malware-demo
cd ~/malware-demo

echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.txt


┌──(root㉿kali)-[~/malware-demo]
└─# ls
eicar.txt

┌──(root㉿kali)-[~/malware-demo]
└─# sha256sum eicar.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267  eicar.txt

┌──(root㉿kali)-[~/malware-demo]
└─# python3 -m http.server 8080

Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
192.168.113.141 - - [31/Jan/2026 13:14:43] "GET /eicar.txt HTTP/1.1" 200 -
192.168.113.141 - - [31/Jan/2026 13:16:51] "GET /eicar.txt HTTP/1.1" 200 -
```

On windows:

Run below commands on PowerShell as Administrator.

- Set-MpPreference -DisableRealtimeMonitoring $false
- netsh advfirewall set allprofiles state on

Verify Network Connectivity

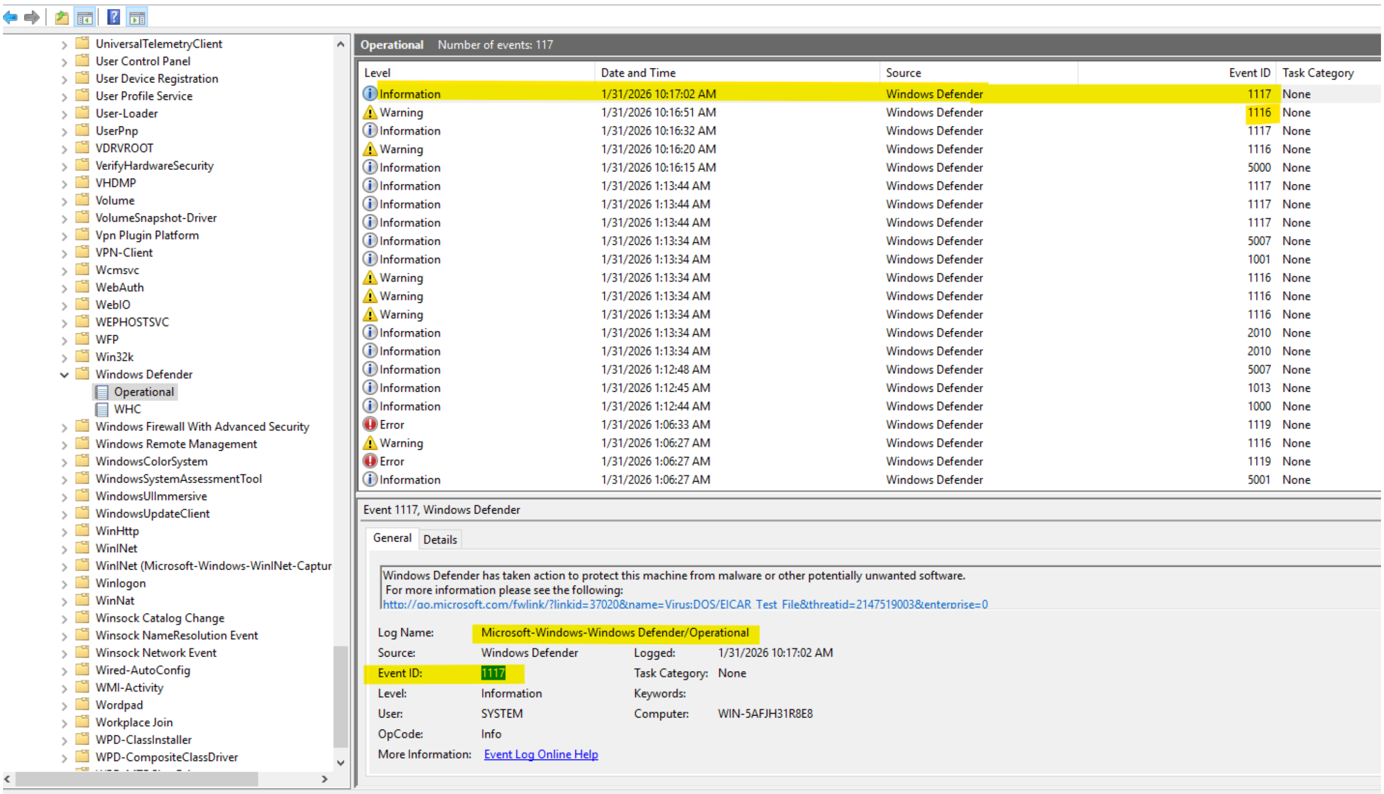
- ping <KALI_IP>

Download the File (Simulated Infection Vector)

- Invoke-WebRequest http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Administrator\Desktop\eicar.txt

Event Logs

Now go to Event Viewer → Application and Service logs → Microsoft → Windows → Windows Defender



Look for:

- Event ID **1116 / 1117**

Capture Traffic:

Run as root on kali:

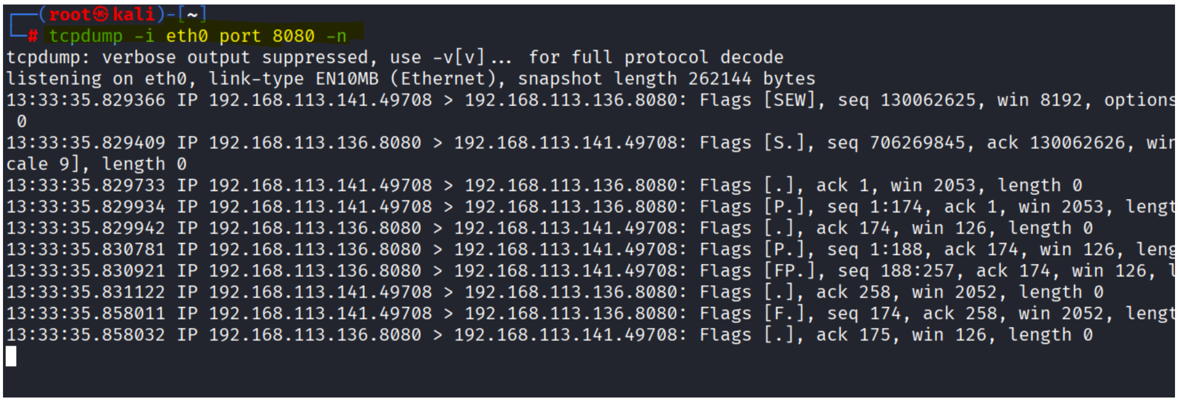
- ➢ tcpdump -i eth0 port 8080 -n

Now, go to window server and rerun the same command:

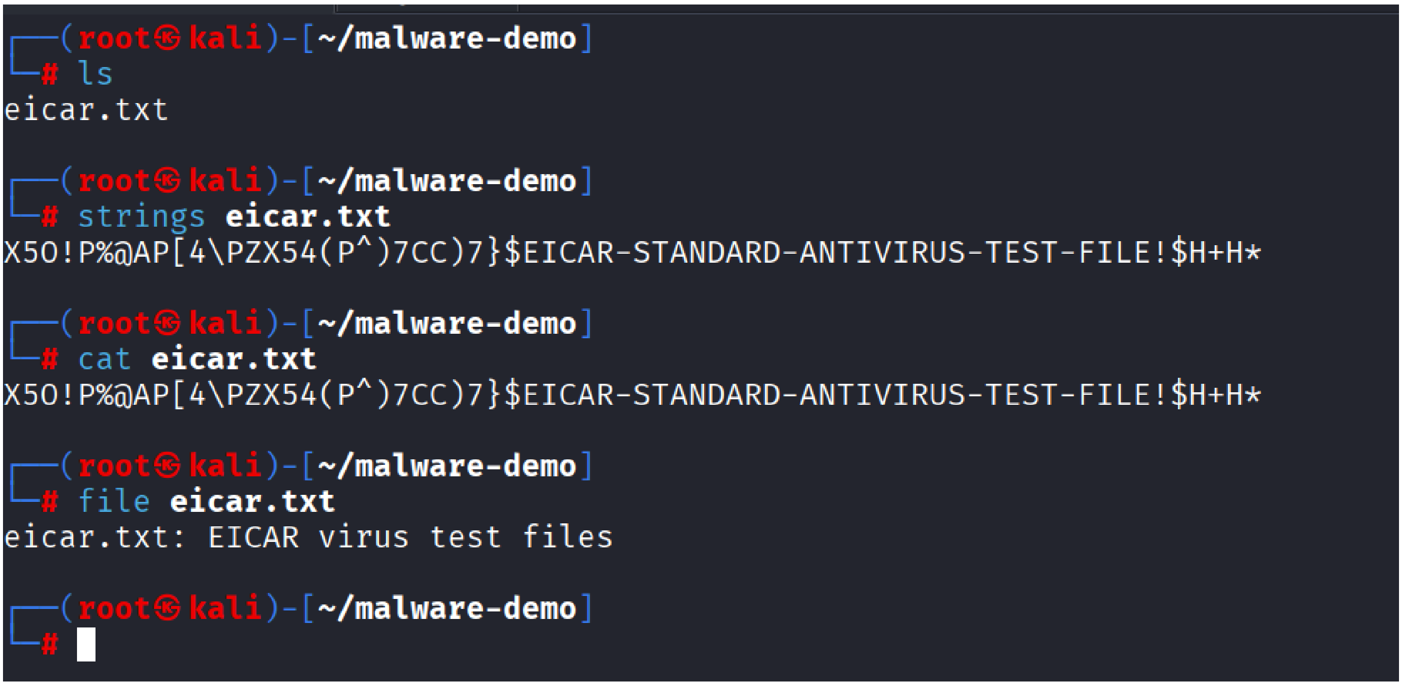
- ➢ Invoke-WebRequest http://<KALI_IP>:8080/eicar.txt -OutFile C:\Users\Public\eicar.txt

Go to kali linux again, rerun the tcpdump command:

- ➢ tcpdump -i eth0 port 8080 -n



Static Analysis demo (on kali)



- Students will learn to create phishing links → done already.